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Concurrent Server TCP/IP

Concurrent Real Time Programming

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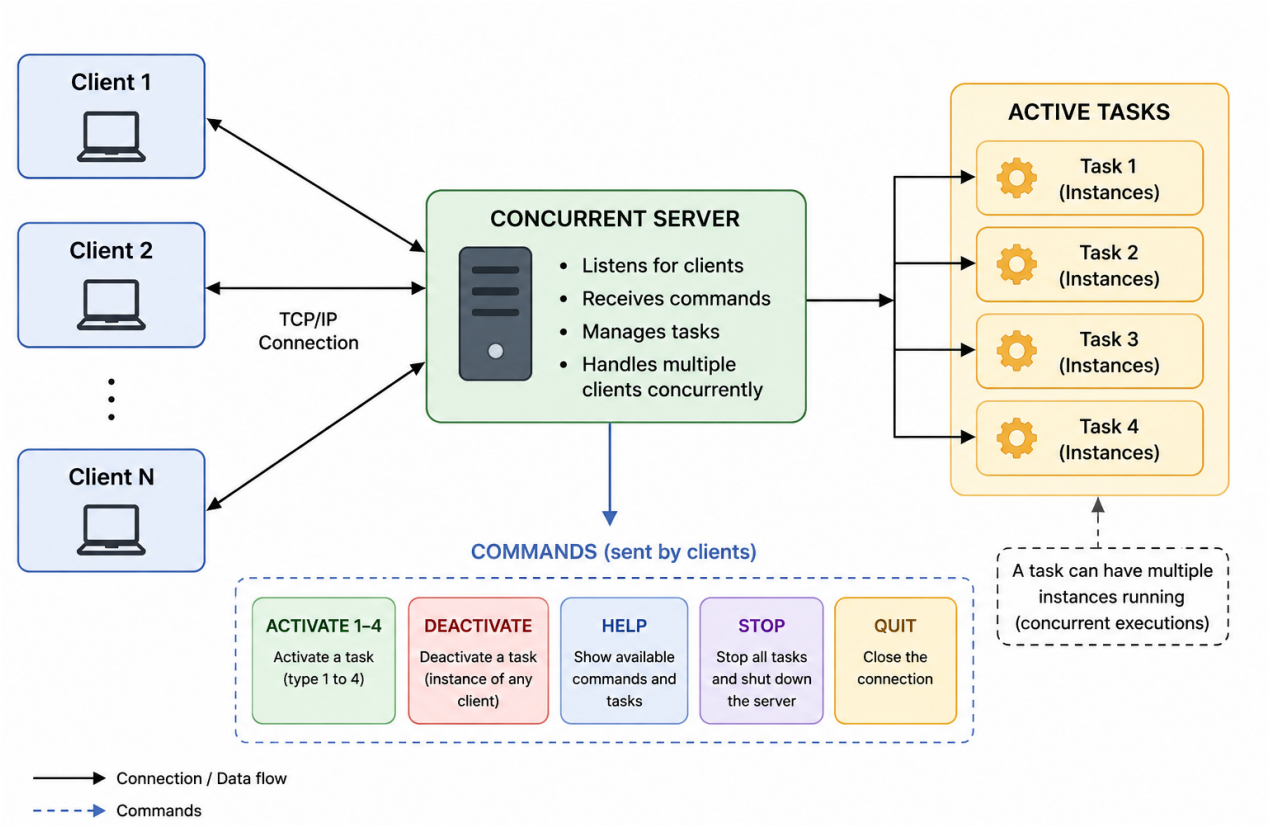
Project Context

- **Objective:** Simulate a dynamic environment where periodic tasks are activated/deactivated via a remote supervisor (Client) through a central controller (Server).
- **System Model:**
  - **Server:** Manages a pool of clients and a pool of real-time worker threads.
  - **Clients:** Issue commands to modify the system state (Activate, Deactivate, Block, Stop, Quit).
  - **Tasks:** Periodic routines with predefined execution times, deadlines and periods.
- **The Challenge:** Managing high-frequency state changes across multiple threads (Connection Handlers, Task Workers, and the Main Server Loop) without incurring race conditions or deadlocks.

System Architecture



- **Decoupled Design:** Separates network connection handling from real-time task execution.
- **1:1 Thread Mapping:** One *Connection Handler* per client socket; one *Worker Thread* per active task instance.

Core Data Structures

To ensure thread safety, the system relies on two primary shared arrays:

1. `Client clients[MAX_THREADS]`:
   - Tracks connected clients, their socket FDs, and assigned ports.
   - **Protected by:** `mutex_clients_array`, `roomAvailable`.
2. `ActiveTask tasks_active[MAX_NUMBER_ACTIVE_TASKS]`:
   - Tracks currently running periodic tasks, their IDs, and thread handles.
   - **Protected by:** `active_tasks_mutex`, `free_slots_sem`.

Note: These structures represent the "Critical Sections" where race conditions are most likely to occur during simultaneous command processing.

Mutexes

The implementation utilizes several `pthread_mutex_t` objects to enforce Mutual Exclusion:

`mutex_clients_array` Protects the integrity of the client registry. Ensures only one thread modifies the client list at a time.

`active_tasks_mutex` Protects the task registry. Critical for preventing corruption when tasks are added or removed.

`mutex_stopserver` Protects the global shutdown flag, preventing a race between the "Stop" command and the main loop's exit condition.

`log_mutex` **Output Integrity:** Essential for real-time debugging. Prevents multiple threads from interleaving their `printf` calls, which would render the server logs unreadable.

Condition Variables

To handle the "Server Full" state for clients, we employ `pthread_cond_t` (`roomAvailable`):

- **The Consumer (Connection Handler):** If `findFreePosition()` returns -1 (Server's array for client is full), the thread enters:
  - `pthread_mutex_lock(&mutex_clients_array)`
  - `pthread_cond_wait(&roomAvailable, &mutex_clients_array)`
- **The Signal:** When `rmvclient()` is called (a client disconnects), `pthread_cond_signal(&roomAvailable)` is invoked, waking up a waiting thread so it can occupy the newly available slot.

Semaphores

The system uses a **Counting Semaphore** (`free_slots_sem`) to manage the `tasks_active` array:

- **Initialization:** `sem_init(&free_slots_sem, 0, MAX_NUMBER_ACTIVE_TASKS)`.
- **Resource Tracking:** The semaphore acts as a counter for available slots.
- **Non-Blocking Allocation:**
  - `sem_trywait` is used during activation.
  - If the semaphore is 0 (no slots), the server immediately rejects the task with a `FULL CAPACITY` message instead of hanging.
- **Reclamation:** When a task is de-activated, `sem_post` increments the counter, allowing the next `ACTIVATE` command to proceed.

Command Logic: ACTIVATE Task

When a client sends an `ACTIVATE <ID>` command, the Server follows this protocol:

1. **Parsing:** Extract task ID from the command string.
2. **Resource Acquisition:**
   - Attempt to decrement the `free_slots_sem`.
   - Lock `active_tasks_mutex`.
   - Locate an empty slot in `tasks_active`.
3. **Schedulability Check:** Run `is_schedulable()`. This is a read-only operation on the task array.
4. **Thread Spawning:** Call `pthread_create` for `handling_active_task`.
5. **Unlock:** Release `active_tasks_mutex`.

Command Logic: BLOCK, QUIT, and STOP

- `BLOCK` **(Deactivate Task):**
  - Finds the oldest task instance in `tasks_active`.
  - Sets field of that task `active = 0` inside a `active_tasks_mutex` block.
  - **Crucial:** The worker thread, seeing `active == 0`, terminates its loop.
  - Calls `sem_post` to signal a slot is now free.
- `QUIT` **(Client Disconnect):**
  - Closes the socket and calls `rmvclient()` which...
  - Uses `mutex_clients_array` to update the client buffer.
- `STOP` **(Global Shutdown):**
  - Updates the global `close_server` flag.
  - **Synchronization:** Uses `mutex_stopserver` to ensure atomic visibility of the shutdown signal to all threads.

Summary of Synchronization Primitives

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Primitive	Variable Name	Synchronization Scope
Mutex	`mutex_clients_array`	Protects Client array state and slots
Mutex	`active_tasks_mutex`	Protects Task array state insertion/del
Mutex	`log_mutex`	Ensures atomic, un-interleaved console
Mutex	`mutex_stopserver`	Prevents race conditions on shutdown
Semaphore	`free_slots_sem`	Non-blocking task capacity tracking (s
Cond Var	`roomAvailable`	Blocks overflow clients until a slot free

Table: Mapping of POSIX primitives to system critical sections.

Conclusion

- **Robustness:** The system successfully bridges the gap between high-level TCP commands and low-level real-time thread management.
- **Concurrency Safety:** By utilizing mutexes for state, semaphores for resources and Condition Variables for flow control.
- **Scalability:** The architecture allows for N clients and M tasks to operate simultaneously, governed strictly by the defined synchronization boundaries.

Future Improvements

- Compare Rate Monotonic with alternative fixed-priority policies.
- Extend evaluation to dynamic-priority algorithms (e.g., EDF).
- Evaluate system performance under high CPU utilization and overloaded conditions.
- Thread pool for clients achitecture.
- Possibly constant scan of all the clients and active tasks